

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department: Computer Science and Engineering

Program: **Bachelor of Science in Computer Science and Engineering**

Semester Final Examination: Spring 2021

Year: **4th** Semester: **1st**

Course Number: **CSE4107**

Course Name: Artificial Intelligence

Time: 3 (Three) hours

Full Marks: 70

Instruction: Use **separate scripts** for **Part A** and **Part B**. Marks allotted are indicated in the right margin.

Part A														
There are 4 (four) questions. Question #1 is compulsory. Answer any 2 (two) from other three.														
Question 1. [Marks: 16 $\frac{2}{3}$]														
a)	Elaborate the concept of Rational Agents.	[3.0]												
b)	Prove or disprove the proposition X using a typical Forward Chaining algorithm on the knowledgebase that follows. A, B, C, D, $G \wedge P \Rightarrow H$, $A \wedge H \Rightarrow E$, $H \wedge K \Rightarrow X$, $E \wedge X \Rightarrow P$, $C \wedge K \Rightarrow L$, $B \wedge F \Rightarrow G$, $B \wedge D \Rightarrow K$, $D \wedge F \Rightarrow G$, $D \wedge G \Rightarrow P$, $D \wedge L \Rightarrow H$.	[4.5]												
c)	Analyze the role of Hill-climbing Local Search strategy in finding reasonable solutions to hard problems with no specific goal states.	[4.5]												
d)	Present a comparative study between Greedy Best-First Search and A* Search algorithms.	[4 $\frac{2}{3}$]												
Question 2. [Marks: 15]														
a)	Show using resolution that the FOL sentence $\exists x(P1(x) \wedge P2("C1", x))$ is true according to the following knowledgebase. i) $\neg P3(y) \vee P2(y, F1(y))$ ii) $P1(F1(y)) \vee \neg P3(y)$ iii) $P3("C1")$	[5.0]												
b)	Perform an analysis regarding use of good Heuristic functions in solving real problems with huge search spaces.	[5.0]												
c)	Evaluate the usefulness of ideas of Evolution in Genetic algorithms for solving optimization problems.	[5.0]												
Question 3. [Marks: 15]														
a)	Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate 'Block'. <table><tr><td>B</td></tr><tr><td>A</td></tr></table><table><tr><td>D</td></tr><tr><td>C</td></tr></table><table><tr><td>E</td></tr><tr><td>F</td></tr></table> <table><tr><td>C</td></tr><tr><td>B</td></tr><tr><td>A</td></tr></table><table><tr><td>F</td></tr><tr><td>E</td></tr><tr><td>D</td></tr></table> Initial StateGoal State	B	A	D	C	E	F	C	B	A	F	E	D	[5.0]
B														
A														
D														
C														
E														
F														
C														
B														
A														
F														
E														
D														

b)	Argue that the action schema is an efficient tool to consider specific features of diverse domains in Planners.	[5.0]																																						
c)	Illustrate the suitability of the MINIMAX search strategy for solving Game Playing problems.	[5.0]																																						
Question 4. [Marks: 15]																																								
a)	Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute $P(d \mid \neg a \wedge b)$ using the network. <table><tr><td>A</td><td>B</td><td>P(c)</td></tr><tr><td>a</td><td>b</td><td>0.9</td></tr><tr><td>a</td><td>$\neg b$</td><td>0.4</td></tr><tr><td>$\neg a$</td><td>b</td><td>0.2</td></tr><tr><td>$\neg a$</td><td>$\neg b$</td><td>0.01</td></tr></table><table><tr><td>A</td><td>C</td><td>P(d)</td></tr><tr><td>a</td><td>c</td><td>0.8</td></tr><tr><td>a</td><td>$\neg c$</td><td>0.5</td></tr><tr><td>$\neg a$</td><td>c</td><td>0.3</td></tr><tr><td>$\neg a$</td><td>$\neg c$</td><td>0.02</td></tr></table><table><tr><td>B</td><td>P(a)</td></tr><tr><td>b</td><td>0.6</td></tr><tr><td>$\neg b$</td><td>0.05</td></tr></table><table><tr><td>P(b)</td></tr><tr><td>0.07</td></tr></table>	A	B	P(c)	a	b	0.9	a	$\neg b$	0.4	$\neg a$	b	0.2	$\neg a$	$\neg b$	0.01	A	C	P(d)	a	c	0.8	a	$\neg c$	0.5	$\neg a$	c	0.3	$\neg a$	$\neg c$	0.02	B	P(a)	b	0.6	$\neg b$	0.05	P(b)	0.07	[5.0]
A	B	P(c)																																						
a	b	0.9																																						
a	$\neg b$	0.4																																						
$\neg a$	b	0.2																																						
$\neg a$	$\neg b$	0.01																																						
A	C	P(d)																																						
a	c	0.8																																						
a	$\neg c$	0.5																																						
$\neg a$	c	0.3																																						
$\neg a$	$\neg c$	0.02																																						
B	P(a)																																							
b	0.6																																							
$\neg b$	0.05																																							
P(b)																																								
0.07																																								
b)	Analyze the advantages and disadvantages of inference using full joint-probability distribution.	[5.0]																																						
c)	Explain how effectively the two components of Decision Theory are incorporated in Decision Networks.	[5.0]																																						

Part B						
There are 3 (three) questions. Answer any 2 (two).						
Question 1. [Marks: 11 $\frac{2}{3}$]						
a)	Explain Oversampling and Undersampling with appropriate examples.				[3.0]	
b)	For a Classification task, from the Confusion Matrix, TP = 23, TN = 24, FN = 32, FP = 31. Calculate Accuracy, Precision, Recall and F Score.				[5.0]	
c)	What are Feature Normalization and Feature Weighting? Explain how these techniques help in Classification approaches with appropriate examples.				[3 $\frac{2}{3}$]	
Question 2. [Marks: 11 $\frac{2}{3}$]						
a)	How does Machine Learning work?				[3.0]	
b)	Consider the following data.				[5.0]	
	Customer	Age	Income	No. of Credit Cards		Class
	George	35	35k	3		Yes
	Rachel	22	50k	1		No
	Steve	63	70k	2		Yes
	Tom	59	40k	2		No
	Anne	25	29k	4		No
	Predict the Class of John whose age is 37, income is 50k and no. of credit cards is 2 using the K Nearest Neighbor Classification approach assuming K = 3.					

c)	What are Single Learner and Ensemble Learner approaches? Explain the significance of Ensemble Learners over Single Learners with appropriate examples.	$[3\frac{2}{3}]$															
Question 3. [Marks: $11\frac{2}{3}$]																	
a)	Explain how Backpropagation works in Artificial Neural Networks.	[3.0]															
b)	Consider the following data. <table border="1"> <thead> <tr> <th>Object</th><th>Attribute 1 (X): weight index</th><th>Attribute 2 (Y): pH value</th></tr> </thead> <tbody> <tr> <td>Medicine A</td><td>1</td><td>1</td></tr> <tr> <td>Medicine B</td><td>2</td><td>1</td></tr> <tr> <td>Medicine C</td><td>4</td><td>3</td></tr> <tr> <td>Medicine D</td><td>5</td><td>4</td></tr> </tbody> </table> Using K Means Clustering technique, partition the objects into two clusters.	Object	Attribute 1 (X): weight index	Attribute 2 (Y): pH value	Medicine A	1	1	Medicine B	2	1	Medicine C	4	3	Medicine D	5	4	[5.0]
Object	Attribute 1 (X): weight index	Attribute 2 (Y): pH value															
Medicine A	1	1															
Medicine B	2	1															
Medicine C	4	3															
Medicine D	5	4															
c)	Consider a document containing 100 words wherein the word <i>tiger</i> appears 3 times. Also, there are 10 million documents, and the word <i>tiger</i> appears in 1000 documents of these. Calculate the TF-IDF for the word <i>tiger</i> .	$[3\frac{2}{3}]$															